

## Problem Sheet 11:

Submission deadline: 12:00, Friday, July 3, 2026; Solutions: Friday, July 10, 2026

### Problem 1 Radiated Power in Thomson Scattering

A plane wave with electric field

$$\mathbf{E} = \text{Re} [e^{-i\omega t} e^{ikx} E_0 \hat{\mathbf{z}}]$$

is incident, in otherwise empty space, on an electron with charge  $-e$  and mass  $m$ . As a result of the motion induced in this way, the electron emits electromagnetic waves. Show that the radiated power per unit solid angle, averaged over one oscillation period, is given by

$$\frac{dP}{d\Omega} = \frac{\mu_0}{32\pi^2 c} \frac{e^4 |E_0|^2}{m^2} \sin^2 \theta$$

*Hints:*

1. Work non-relativistically, so that you can use the approximations of the Lorentz model and, in particular, the resulting formula for the induced electric dipole moment.
2. Start from the dipole radiation formula

$$\left( \frac{dP}{d\Omega} \right)_{El} = \frac{\mu_0}{16\pi^2 c} |\hat{\mathbf{r}} \times \ddot{\mathbf{p}}_{\text{ret}}|^2 \quad (1)$$

Note that this formula is initially not time-averaged and that  $\mathbf{p}_{\text{ret}}$  denotes the retarded electric dipole moment.

### Solution:

Let the electron be at the origin. Then the incident electric field at the position of the electron is

$$\mathbf{E}(0, t) = \text{Re} (E_0 e^{-i\omega t} \hat{\mathbf{z}}).$$

We work non-relativistically and neglect the effect of the magnetic field on the electron. Furthermore, let the wavelength of the electric field be much larger than the typical amplitude of the electron motion.

Then the equation of motion is

$$m\ddot{\mathbf{r}}(t) = -e \mathbf{E}(0, t).$$

Thus we obtain

$$\ddot{\mathbf{p}}(t) = \frac{e^2}{m} \mathbf{E}(0, t).$$

With

$$\mathbf{p}(t) = \text{Re} \left( \mathbf{p}_0 e^{-i\omega t} \right)$$

it follows that

$$\mathbf{p}_0 = -\frac{e^2}{m\omega^2} E_0 \hat{\mathbf{z}}.$$

With the given dipole radiation formula

$$\frac{dP}{d\Omega} = \frac{\mu_0}{16\pi^2 c} |\hat{\mathbf{r}} \times \ddot{\mathbf{p}}_{\text{ret}}|^2$$

we have

$$\ddot{\mathbf{p}}_{\text{ret}}(t) = \ddot{\mathbf{p}} \left( t - \frac{r}{c} \right).$$

Therefore

$$\ddot{\mathbf{p}}_{\text{ret}} = \text{Re} \left[ \frac{e^2}{m} E_0 e^{-i\omega \left( t - \frac{r}{c} \right)} \hat{\mathbf{z}} \right].$$

We have

$$|\hat{\mathbf{r}} \times \hat{\mathbf{z}}| = \sin \theta.$$

Thus the instantaneous radiated power per unit solid angle is

$$\frac{dP(t)}{d\Omega} = \frac{\mu_0}{16\pi^2 c} \frac{e^4}{m^2} \left[ \text{Re} \left( E_0 e^{-i\omega \left( t - \frac{r}{c} \right)} \right) \right]^2 \sin^2 \theta.$$

Averaging over

$$T = \frac{2\pi}{\omega}$$

gives a factor

$$\frac{1}{2}.$$

Thus it follows that

$$\left\langle \frac{dP}{d\Omega} \right\rangle = \frac{\mu_0}{32\pi^2 c} \frac{e^4}{m^2} |E_0|^2 \sin^2 \theta.$$

## Problem 2 The Drude Model

(2.a) Let the half-space  $z \geq 0$  be filled with a conducting medium with  $\mu = \mu_0$ , which is well described by the Drude model, equations

$$\frac{\epsilon(\omega)}{\epsilon_0} \approx 1 + i \frac{Ne^2}{\epsilon_0 m \omega \gamma}, \quad (2)$$

$$\sigma(\omega) \approx \frac{Ne^2}{m\gamma}, \quad (3)$$

described. A plane wave of frequency

$$\epsilon_0 \omega \ll \sigma$$

is normally incident on the medium from the vacuum region  $z < 0$ . Thus the incident wave has the form

$$\mathbf{A}_I = \mathbf{C}_I e^{-i\omega t} e^{ikz},$$

the reflected wave has the form

$$\mathbf{A}_R = \mathbf{C}_R e^{-i\omega t} e^{-ikz},$$

and the transmitted wave has the form

$$\mathbf{A}_T = \mathbf{C}_T e^{-i\omega t} e^{-(1-i)z/\delta}.$$

Determine  $\mathbf{C}_R$  and  $\mathbf{C}_T$  in terms of  $\mathbf{C}_I$ ,  $\omega$ , and  $\sigma$ . Show that the Poynting flux of the reflected wave is smaller than the Poynting flux of the incident wave.

- (2.b) Consider the same setup as in part (a), but now replace the conducting medium described by equation (2) by a plasma described by the equation

$$\frac{\epsilon(\omega)}{\epsilon_0} \approx 1 - \frac{\omega_p^2}{\omega^2} \quad (4)$$

with plasma frequency  $\omega_p^2 = \frac{Ne^2}{\epsilon_0 m}$ . A plane wave

$$\mathbf{A}_I = \mathbf{C}_I e^{-i\omega t} e^{ikz}$$

with

$$\omega < \omega_p$$

is incident from the region  $z < 0$ . Determine the reflected and transmitted waves. Show that the Poynting flux of the reflected wave is equal to the Poynting flux of the incident wave.

### Solution:

We choose

$$\mathbf{A}(z, t) = A(z, t) \hat{\mathbf{x}}.$$

We work in a gauge in which

$$\mathbf{E} = -\partial_t \mathbf{A} = i\omega A \hat{\mathbf{x}}, \quad \mathbf{B} = \nabla \times \mathbf{A} = \partial_z A \hat{\mathbf{y}}.$$

The tangential components of  $\mathbf{E}$  and  $\mathbf{H}$  must be continuous at the interface. This gives

$$A(0^-, t) = A(0^+, t), \quad \partial_z A(0^-, t) = \partial_z A(0^+, t).$$

- (2.a) We have

$$\epsilon(\omega) = \epsilon_0 + \frac{i\sigma}{\omega}, \quad \sigma = \frac{Ne^2}{m\gamma}, \quad \sigma \gg \epsilon_0 \omega.$$

The dispersion relation in the metal is

$$k^2 = \mu_0 \epsilon(\omega) \omega^2 \simeq i\mu_0 \sigma \omega,$$

since

$$1 \ll \frac{\sigma}{\epsilon_0 \omega}.$$

With the skin depth

$$\delta = \sqrt{\frac{2}{\mu_0 \sigma \omega}}$$

we write the transmitted wave as

$$\mathbf{A}_T = \mathbf{C}_T e^{-i\omega t} e^{-(1-i)z/\delta}.$$

The boundary conditions at  $z = 0$  give

$$\mathbf{C}_I + \mathbf{C}_R = \mathbf{C}_T$$

as well as

$$ik(\mathbf{C}_I - \mathbf{C}_R) = -\frac{1-i}{\delta} \mathbf{C}_T.$$

Hence

$$ik(\mathbf{C}_I - \mathbf{C}_R) = -\frac{1-i}{\delta} (\mathbf{C}_I + \mathbf{C}_R).$$

This yields

$$\mathbf{C}_R = \frac{ik + \frac{1-i}{\delta}}{ik - \frac{1-i}{\delta}} \mathbf{C}_I$$

and

$$\mathbf{C}_T = \frac{2ik}{ik - \frac{1-i}{\delta}} \mathbf{C}_I.$$

Setting

$$\alpha = k\delta > 0,$$

gives

$$\mathbf{C}_R = \frac{i\alpha + (1-i)}{i\alpha - (1-i)} \mathbf{C}_I.$$

We define the reflection coefficient

$$r := \left| \frac{\mathbf{C}_R}{\mathbf{C}_I} \right|.$$

Then

$$r = \left| \frac{i\alpha + (1-i)}{i\alpha - (1-i)} \right| = \sqrt{\frac{(\alpha-1)^2 + 1}{(\alpha+1)^2 + 1}} < 1 \quad \text{for } \alpha > 0.$$

Thus

$$|\mathbf{C}_R|^2 < |\mathbf{C}_I|^2.$$

The time-averaged Poynting flux of the incoming and reflected wave is proportional to the amplitude  $|\mathbf{C}|^2$ . Therefore

$$\frac{|\langle \mathbf{S}_R \rangle|}{|\langle \mathbf{S}_I \rangle|} = \frac{|\mathbf{C}_R|^2}{|\mathbf{C}_I|^2} = r^2 = \frac{(\alpha-1)^2 + 1}{(\alpha+1)^2 + 1} < 1.$$

(2.b) Now

$$\frac{\varepsilon(\omega)}{\varepsilon_0} \simeq 1 - \frac{\omega_p^2}{\omega^2}.$$

The dispersion relation is

$$k_p^2 = \mu_0 \varepsilon(\omega) \omega^2 = \frac{\omega^2 - \omega_p^2}{c^2}.$$

For

$$\omega < \omega_p$$

we have

$$k_p^2 < 0$$

and hence  $k_p$  is imaginary. We define

$$\kappa = \frac{1}{c} \sqrt{\omega_p^2 - \omega^2}.$$

Thus

$$k_p = i\kappa.$$

The transmitted wave is therefore the evanescent wave

$$\mathbf{A}_T = \mathbf{C}_T e^{-i\omega t} e^{-\kappa z}.$$

A wave proportional to  $e^{\kappa z}$  would formally also be a solution, but it is not physically meaningful for  $z \rightarrow \infty$ .

The boundary conditions at  $z = 0$  again give

$$\mathbf{C}_I + \mathbf{C}_R = \mathbf{C}_T$$

and

$$ik(\mathbf{C}_I - \mathbf{C}_R) = -\kappa \mathbf{C}_T.$$

With

$$r = \frac{\mathbf{C}_R}{\mathbf{C}_I}, \quad t = \frac{\mathbf{C}_T}{\mathbf{C}_I}$$

it follows that

$$1 + r = t$$

and

$$ik(1 - r) = -\kappa t.$$

This yields

$$r = \frac{ik + \kappa}{ik - \kappa}$$

and

$$t = \frac{2ik}{ik - \kappa}.$$

Now

$$|r| = \sqrt{\frac{k^2 + \kappa^2}{k^2 + \kappa^2}} = 1.$$

The intensity of the incident wave is therefore completely reflected. The time-averaged energy flux of the evanescent wave in the  $z$ -direction vanishes.

### Problem 3 Fresnel Problem for a Topological Insulator

The optical properties of a remarkable class of materials, the so-called *topological insulators* (TIs), are described by constitutive relations in which the fine-structure constant

$$\alpha = \frac{e^2/\hbar c}{4\pi\epsilon_0}$$

appears. With

$$\alpha_0 = \alpha \sqrt{\frac{\epsilon_0}{\mu_0}}$$

the constitutive relations are

$$\mathbf{D} = \epsilon \mathbf{E} - \alpha_0 \mathbf{B}, \quad \mathbf{H} = \frac{\mathbf{B}}{\mu} + \alpha_0 \mathbf{E}.$$

- (3.a) Start from Maxwell's equations in matter without free charges or free currents. Show that a monochromatic plane wave of the fields  $\mathbf{E}$  and  $\mathbf{B}$  is a solution of these equations for a topological insulator, and determine the wave velocity.
- (3.b) A linearly polarized plane wave is normally incident on the planar surface of a topological insulator. Show that the transmitted wave remains linearly polarized, with its electric field rotated by an angle  $\theta_F$ . This is called the *Faraday rotation* of the polarization plane.  
*Hint: Choose the incident electric field such that it is polarized in the  $x$ -direction:*

$$\mathbf{E}_i = E_i \hat{\mathbf{x}} e^{i(k_0 z - \omega t)}.$$

The reflected wave propagates in the  $-z$ -direction, while the transmitted wave propagates in the  $+z$ -direction. We therefore write

$$\mathbf{E}_r = E_i (r_x \hat{\mathbf{x}} + r_y \hat{\mathbf{y}}) e^{i(-k_0 z - \omega t)},$$

and

$$\mathbf{E}_t = E_i (t_x \hat{\mathbf{x}} + t_y \hat{\mathbf{y}}) e^{i(kz - \omega t)}.$$

Then use the appropriate continuity conditions of the fields at the interface  $z = 0$  to establish a relation between  $t_x$  and  $t_y$  and finally determine  $\theta_F = \arctan(t_y/t_x)$ .

- (3.c) Show that the reflected wave remains linearly polarized, with its electric field rotated by an angle  $\theta_K$ . This is called the *Kerr rotation* of the polarization plane.  
*Hint: Proceed as in (3.b) to determine the angle  $\theta_K = \arctan(r_y/r_x)$  analogously.*

### Solution:

- (3.a) For plane monochromatic waves with  $\mathbf{E} = \mathbf{C}_E \exp(i\mathbf{k}\mathbf{r} - i\omega t)$  and  $\mathbf{B} = \mathbf{C}_B \exp(i\mathbf{k}\mathbf{r} - i\omega t)$ , the four Maxwell equations without free currents and charges are

$$\mathbf{k} \cdot \mathbf{D} = 0 \tag{5}$$

$$\mathbf{k} \cdot \mathbf{B} = 0 \tag{6}$$

$$\mathbf{k} \times \mathbf{E} = \omega \mathbf{B} \tag{7}$$

$$\mathbf{k} \times \mathbf{H} = -\omega \mathbf{D} \tag{8}$$

Inserting the constitutive relations, the first Maxwell equation gives

$$\mathbf{k} \cdot \mathbf{D} = \epsilon \mathbf{k} \cdot \mathbf{E} - \alpha_0 \mathbf{k} \cdot \mathbf{B} \Rightarrow \mathbf{k} \cdot \mathbf{E} = 0 \quad (9)$$

The fourth equation gives

$$\mathbf{k} \times \mathbf{H} = \frac{1}{\mu} \mathbf{k} \times \mathbf{B} + \alpha_0 \mathbf{k} \times \mathbf{E} = -\omega(\epsilon \mathbf{E} - \alpha_0 \mathbf{B}) \Rightarrow \mathbf{k} \times \mathbf{B} = -\omega \epsilon \mu \mathbf{E}. \quad (10)$$

These are the usual wave equations for electromagnetic waves in vacuum, but with  $\epsilon_0 \mu_0 \rightarrow \epsilon \mu$ . We therefore obtain wave solutions with wave velocity

$$v = \frac{1}{\sqrt{\epsilon \mu}}. \quad (11)$$

(3.b) Since the wave is normally incident on the planar surface, the fields  $\mathbf{E}$  and  $\mathbf{B}$  are transverse to the surface, and we obtain the two boundary conditions

$$\mathbf{E}_i + \mathbf{E}_r = \mathbf{E}_t. \quad (12)$$

In components we obtain

$$1 + r_x = t_x \quad (13)$$

$$r_y = t_y. \quad (14)$$

The second boundary condition is

$$\mathbf{H}_i + \mathbf{H}_r = \mathbf{H}_t \quad (15)$$

$$\Rightarrow \frac{1}{\mu_0} (\mathbf{B}_i + \mathbf{B}_r) = \frac{1}{\mu} \mathbf{B}_t + \alpha_0 \mathbf{E}_t \quad (16)$$

$$\Rightarrow \underbrace{\frac{1}{\mu_0} \frac{k_0}{\omega_0}}_{=\sqrt{\frac{\epsilon_0}{\mu_0}}} \hat{\mathbf{z}} \times (\mathbf{E}_i - \mathbf{E}_r) = \underbrace{\frac{1}{\mu} \frac{k}{\omega}}_{=\sqrt{\frac{\epsilon}{\mu}}} \hat{\mathbf{z}} \times \mathbf{E}_t + \alpha_0 \mathbf{E}_t, \quad (17)$$

where in the last line we used  $\mathbf{B} = \frac{1}{\omega} \mathbf{k} \times \mathbf{E}$ . Thus, in components,

$$\sqrt{\frac{\epsilon_0}{\mu_0}} (1 - r_x) = \sqrt{\frac{\epsilon}{\mu}} t_x + \alpha_0 t_y \quad (18)$$

$$\sqrt{\frac{\epsilon_0}{\mu_0}} r_y = \sqrt{\frac{\epsilon}{\mu}} t_y - \alpha_0 t_x \quad (19)$$

$$(20)$$

To compute  $\theta_F$ , we now insert (14) into (19).

$$\sqrt{\frac{\epsilon_0}{\mu_0}} t_y = \sqrt{\frac{\epsilon}{\mu}} t_y - \alpha_0 t_x \quad (21)$$

$$\Rightarrow \frac{t_y}{t_x} = \frac{\alpha_0}{\sqrt{\epsilon/\mu} + \sqrt{\epsilon_0/\mu_0}}. \quad (22)$$

(3.c) Here we first divide equations (18) and (19) by  $r_x$ .

$$\sqrt{\frac{\epsilon_0}{\mu_0}} \left( \frac{1}{r_x} - 1 \right) = \sqrt{\frac{\epsilon}{\mu}} \frac{t_x}{r_x} + \alpha_0 \frac{t_y}{r_x} \quad (23)$$

$$\sqrt{\frac{\epsilon_0}{\mu_0}} \frac{r_y/r_x}{r_x} = \sqrt{\frac{\epsilon}{\mu}} \frac{t_y}{r_x} - \alpha_0 \frac{t_x}{r_x} \quad (24)$$

$$(25)$$

Next we use Eq. (13) and write

$$\frac{1}{r_x} - 1 = \frac{t_x - r_x}{r_x} - 1 = \frac{t_x}{r_x} - 2. \quad (26)$$

Inserting this and also using (14) to replace  $t_y/r_x = r_x/r_x$ , we obtain

$$\sqrt{\frac{\epsilon_0}{\mu_0}} \left( \frac{t_x}{r_x} - 2 \right) = \sqrt{\frac{\epsilon}{\mu}} \frac{t_x}{r_x} + \alpha_0 \frac{r_y}{r_x} \quad (27)$$

$$\sqrt{\frac{\epsilon_0}{\mu_0}} \frac{r_y}{r_x} = \sqrt{\frac{\epsilon}{\mu}} \frac{r_y}{r_x} - \alpha_0 \frac{t_x}{r_x} \quad (28)$$

$$(29)$$

This is now a system of two equations with two unknowns that we can solve in the standard way. The last equation gives

$$\frac{t_x}{r_x} = \frac{1}{\alpha_0} \left( \sqrt{\frac{\epsilon}{\mu}} + \sqrt{\frac{\epsilon_0}{\mu_0}} \right) \frac{r_y}{r_x}. \quad (30)$$

Inserting this into the first equation, we obtain

$$\alpha_0 \frac{r_y}{r_x} + \left( \sqrt{\frac{\epsilon}{\mu}} - \sqrt{\frac{\epsilon_0}{\mu_0}} \right) \frac{t_x}{r_x} = -2 \sqrt{\frac{\epsilon_0}{\mu_0}} \quad (31)$$

$$\Rightarrow \left[ \alpha_0 + \frac{1}{\alpha_0} \left( \frac{\epsilon}{\mu} - \frac{\epsilon_0}{\mu_0} \right) \right] \frac{r_y}{r_x} = -2 \sqrt{\frac{\epsilon_0}{\mu_0}} \quad (32)$$

$$\Rightarrow \frac{r_y}{r_x} = \frac{-2\alpha_0 \sqrt{\frac{\epsilon_0}{\mu_0}}}{\alpha_0^2 + \left( \frac{\epsilon}{\mu} - \frac{\epsilon_0}{\mu_0} \right)}. \quad (33)$$